

Reader Report: Week 06

Probabilistic Invariance: VI-BIP and the Core Theorems

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Summary

Wu and Blei (2025) place a prior over binary selectors $z \in \{0, 1\}^p$ indexing the invariant features; the posterior $p(z \mid \mathcal{D}) \propto p(z) \prod_{e,i} \hat{g}(Y_i^e \mid X_{ei}^z) / \hat{p}_e(Y_i^e \mid X_{ei}^z)$ rewards selectors for which the pooled conditional $\hat{g}$ dominates each local $\hat{p}_e$ —a probabilistic lift of Peters et al.’s test. Consistency (Thm 1) requires $\mu_{\min} > 0$; contraction (Thm 2) gives mass on $\{z \neq z^*\}$ decaying as $O(Re^{-\kappa n E \mu_{\min}})$. VI-BIP approximates the posterior by mean-field ELBO maximization, using the U2G estimator to handle discrete gradients; it recovers the same invariant gene sets as exact BIP on a 6,170-gene dataset. Fan et al. (2024) take a different tack—a minimax formulation with negative environment weights—returning a point estimate $\hat{\beta}^\gamma$ with polynomial-time rates but no uncertainty.

Critique

The contraction exponent $\kappa n E \mu_{\min}$ is more informative than it looks. Rearranging the bound to achieve ϵ posterior mass on bad selectors requires $nE \gtrsim \frac{p \log 2 + \log(1/\epsilon)}{\kappa \mu_{\min}}$, i.e. roughly $O(p/(\mu_{\min} E))$ samples per environment. This makes explicit a tension the paper leaves implicit: $\mu_{\min}$ is a property of the *intervention distribution*, not the sample, so collecting more data within a given set of environments hits a wall once n is large but E is fixed. For my NIMBYism project with $K \approx 4$ regimes, the bound says I need $\mu_{\min}$ to be large—meaning the policy shifts (parking reform, HOME Initiative) must genuinely alter the distribution of non-causal features—and that I cannot compensate for weak interventions purely by accumulating more census tracts.

Against NegDRO, the difference is not just uncertainty vs. point estimate. NegDRO’s negative weights can *amplify* contrasts between specific environment pairs, effectively up-weighting interventions that most violate invariance. BIP, by contrast, treats environments symmetrically inside the product likelihood; it cannot selectively stress-test a single environment pair. This means NegDRO may be more powerful when one environment is known *a priori* to be the most informative intervention, while BIP is preferable when intervention quality is unknown and one wants the data to speak.

VI-BIP’s mean-field family $\prod_j \text{Bern}(\pi_j)$ ignores posterior correlation between z_j and z_k . For correlated covariates—say, school quality and income in census tracts—the true posterior likely assigns mass jointly to $\{z_j = z_k = 1\}$ or $\{z_j = z_k = 0\}$ rather than independently. Mean-field will spread this mass, biasing each π_j toward 0.5 and understating our confidence in either covariate being causal.

Questions

1. The sample complexity argument above suggests a diagnostic: estimate $\hat{\mu}_e$ for each environment pair, flag pairs where it is near zero as “weak interventions,” and prioritize collecting new environments there rather than more data within existing ones. Has this been explored as a practical acquisition strategy within the BIP framework?
2. Can the U2G estimator be paired with a structured (e.g. Ising) variational family to handle correlated selectors, or does U2G’s Gaussian relaxation fundamentally assume independence across z_j ?

References

- Y. Fan, S. Li, and Z. Yang. Causal invariance learning via efficient nonconvex optimization. *arXiv preprint arXiv:2412.11850*, 2024. URL <https://arxiv.org/abs/2412.11850>.
- Y. Wu and D. M. Blei. Bayesian invariance modeling of multi-environment data. *arXiv preprint arXiv:2506.22675*, 2025. URL <https://arxiv.org/abs/2506.22675>.